

Spatiotemporal Perception of Single Overlapped Vibrotactile Stimulation to Multiple Body Locations

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Abstract—Haptic information is useful in various fields, and presenting stimulation to multiple points enriches the experience. Independent Vibrotactile Stimulation to multiple points (IVS) evokes a tactile apparent motion. We propose Single Overlapped Vibrotactile Stimulation (SOVS) with imagination to produce spatiotemporal perception like IVS. The SOVS combines the stimulation of IVS. We developed a pseudo-dribbling system that produces a feeling of an imaginary ball moving between the hand and the feet where the same stimulation was given. To examine the feeling by SOVS, the first experiment on changing the speed of the stimulation showed the SOVS can produce a similar feeling to the IVS, indicating SOVS allows participants to match the perceived stimulation to the stimulation according to the dribbling being imagined. The second experiment was conducted using an obstacle between the hand and the feet to interrupt the image. We controlled the interruption by controlling attention. Results showed that both IVS and SOVS overcome the obstacle to evoke a piercing feeling under the attention to the floor, whereas IVS still provided a piercing feeling under the attention to the obstacle but SOVS could not provide it, suggesting the possibility that the perception mechanism is different between the SOVS and IVS.

Index Terms—Vibrotactile Stimulation, Apparent Motion, Single Overlapped Vibrotactile Stimulation, Tactile Perception

I. INTRODUCTION

Haptic information is closely related to a sense of presence, improves the sense of immersion, and provides realistic feelings to virtual space and objects. As haptic information is relevant to physical experiences, technologies for presenting haptic senses are useful in various fields including manufacturing, education, medical, welfare, and entertainment [1] [2].

The basic approach to developing a haptic experience is to present a stimulation based on the perception principle

[3] [4]. A perception principle provides a simple design for a haptic display owing to the dominant cue for perception; however, the perceptual principle findings are necessary for each perception. Cross- or multi-modal approaches are also useful for presenting haptic sensations. A tactile sensation can be derived by visual stimulation even without any haptic stimulations; in such cases, is biased by visual [5] and/or auditory senses [6].

To enrich the haptic experience, presenting vibrotactile stimuli to multiple body locations [7] [8] is a well-used method because it allows the users to feel the spatiotemporal changes of the stimulation. Moreover, stimulation to multiple points can generate a phantom sensation [9], i.e., a virtual stimulus perceived between two stimuli given simultaneously, and the tactile apparent motion [10], i.e., a feeling that the stimulus is moving between two points stimulated in sequence. The phantom sensation is also reported to occur along large body sites [11] and in and out of the body [12]. Both phantom sensation and tactile apparent motion are hypothetically increasing the number of actuators to generate a sensation to a location with no actuator which is enriching the haptic experience. Although presenting different stimulations to multiple body locations is capable of improving the haptic experience, the required number of actuators and controllers for time synchronization also increases.

Another way to improve the haptic experience is to provide context to users along with the stimulation. For example, presenting vibrations with vibrotactile actuators to users holding an empty cup [13] or a pole while moving it [14] with vibrotactile actuators equipped leads to a sensation of a ball moving inside. This phenomenon indicates that the context can influence the perception of an object involving vibrotactile stimulation. We can also observe the influences of individual experiences on the perception of visual information. In “the dress” photo posted on the internet on February 2015, different observers spontaneously came to different recognitions even while observing on the same screen [15]. It was considered that individual

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experiences led to different first impressions of the color of the dress.

From these different approaches, we believe people are capable of interpreting ambiguous haptic stimulations by presenting stimuli to multiple points, using cross- or multi-modal stimuli, or being provided a context, and also that our spatiotemporal perception is transformable. For this paper, we propose using synthetic vibrotactile stimulation (SOVS) to present the stimulations by mixing multiple independent multiple stimuli to provide an expected spatiotemporal perception. This approach simplifies the control of the stimulation and eases the presentation of stimuli to multiple points by reducing the needed channels and computation. Komazaki et al. [16] used the collected audio from a professional fencing game and presented it as a vibrotactile stimulus to the hand and the feet through a sabre-imitated device and a piste-imitated device. They reported that the participants were able to feel as if they were participating in the game as the player they were imitating although presented with the same stimulation. From this report, we hypothesized that stimuli that have multiple information combined are capable of providing multiple spatiotemporal perceptions as well. There should be many limitations to this hypothesis, and we mainly considered 2 factors to select the experience. The first factor is that the experience must be provided with multiple vibrotactile information along different body locations. The second factor is that the time difference between the stimuli when combined together isn't too short or long.

Thus for this paper, we will focus on the experience of dribbling a basketball as shown in Fig. 1. We hypothesize that presenting the SOVS while holding an image of the movement can induce a similar feeling as the tactile apparent motion perceived when presenting independent vibrotactile stimulation to multiple points (IVS). To substantiate this hypothesis, we conducted two experiments on the haptic presentation for the selected experience. For both experiments, the hand and the feet were given vibrotactile stimulation, which was measured while the participant was dribbling a basketball. As shown in Fig. 2, we prepared two different methods for stimulation: IVS (conventional method) and SOVS (proposed method). It is expected that the IVS evokes the tactile apparent motion. The two experiments were conducted to investigate the influences of the realism of the stimulation and contributions of the imagination in the SOVS, as compared with the illusion of the tactile apparent motion induced by IVS. In the first experiment, the speed of the tactile stimulation was changed. In the second experiment, as controlling the imagination of each user is difficult, we used an obstacle to obscure the participant's feet and guided their sight. Even when the obstacle was inserted between the hand and the foot, we considered from two iterations of differential stimulation might induce a tactile apparent motion between them [20]. Compared with IVS, we tested whether presenting SOVS and holding the image in the

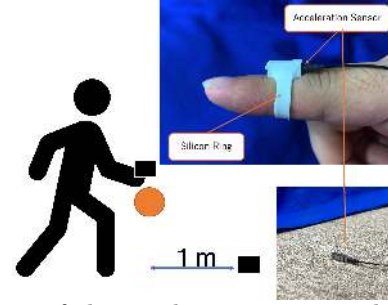


Fig. 1: Places of the acceleration sensor when measuring the vibrations: an acceleration sensor was mounted to the floor 1m away from the dribbler and the index finger of the dribbler

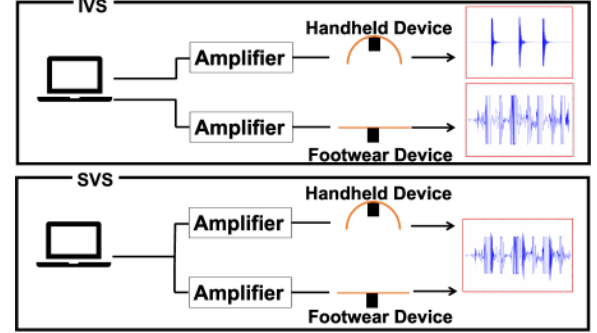


Fig. 2: Prepared stimulations for the experiment: IVS presents two different vibrotactile stimulations from each device, SOVS presents the same combined vibrotactile stimulation from both devices

imagination could overcome the obstacle, i.e., by inducing a similar feeling of the tactile apparent motion.

II. STIMULATION AND SETUP

For the two experiments, we built a system where we could experience dribbling while using vibrotactile stimulation. The stimulation was measured while dribbling the basketball using an acceleration sensor (Showa Sokki Corporation, Vibration Pickup model-2303B). As shown in Fig. 1, we used two sensors, one mounted on the floor and the other mounted on the index finger with a silicon rubber ring to measure the vibrations provided to our feet and hands respectively. We recorded the vibration and the appearance of dribbling by using OBS (OBS Studio Contributors) and created the SOVS using Audition (Adobe). The measured vibration is shown in Fig. 3 and each dribble took around 0.7s for the ball to come back and forth.

The stimulation presenting system consisted of two vibrotactile actuators (Acouve Lab. VP416) (Foster,639897). Each device was used for footwear and handheld devices. The footwear device imitated the structure of the floor of the gym floor and the handheld device imitated the shape of the basketball as shown in Fig. 4. The dimensions of the footwear device were sufficiently large to be able to support the weight of

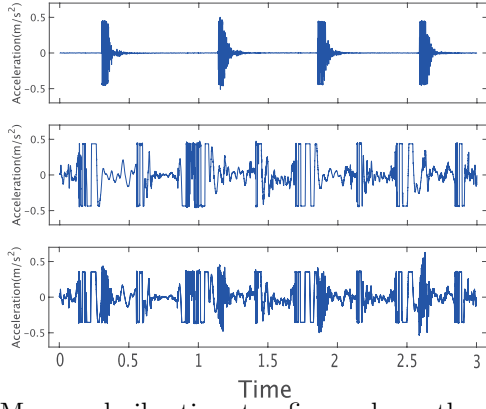


Fig. 3: Measured vibration: top figure shows the vibration from the floor, middle figure shows the vibration from the index finger, bottom figure shows the SOVS of the two vibrations

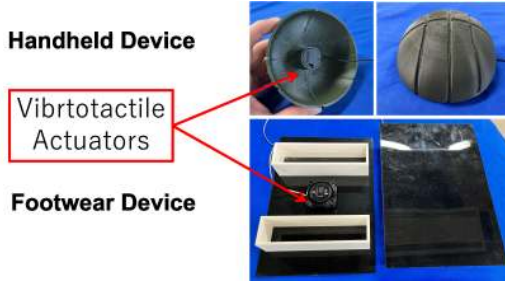


Fig. 4: Developed Devices: top 2 pictures show the handheld device that has a vibrotactile actuator mounted inside, the bottom picture shows the footwear device with a vibrotactile actuator mounted on the backside



Fig. 5: Experimental state: the left state is for Experiment 1, the right state is for Experiment 2 where an obstacle is installed between the handheld device and footwear device

an adult male. The shape of the handheld device was a hemisphere with a diameter of 110mm, ie., sufficiently small that most people would be able to hold it one-handed. Fig. 5 shows the state when the users were using the devices, the participants step on the footwear device and hold the handheld device to stimulate both feet and the dominant hand.

The vibrations presented on each device were measured

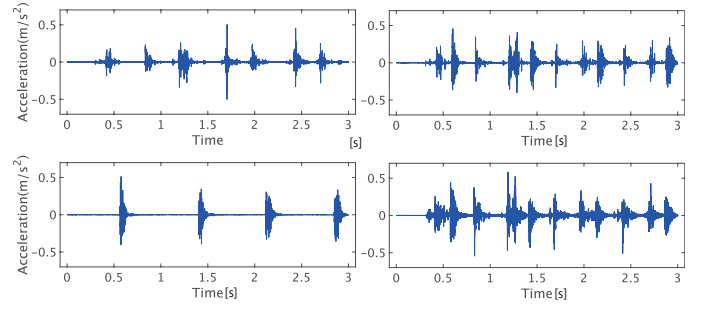


Fig. 6: Reproduced vibration: The top 2 figures are the measured vibration produced on the handheld device, the bottom 2 figures are the vibration produced on the footwear device, and the left figures are the vibration from IVS and the right figures are the vibration from SOVS

using an acceleration sensor on them. Fig. 6 and Fig. ?? show the vibration measured when presenting the IVS or SOVS on each device, respectively. The top waves in each figure show the vibration measured on the handheld device and the bottom waves show the measured vibration from the footwear device. It can be seen that the vibrotactile stimulations collected on the hand and foot are mostly similar owing to the SOVS.

III. EXPERIMENT 1: THE EFFICIENCY OF SYNTHETIC VIBROTACTILE STIMULATION (SOVS)

The first experiment aimed to investigate the generation of a movement feeling like the tactile apparent movement for SOVS compared with IVS, and the effect of the reality of the stimulation (by changing the frequency of the stimulation).

A. Method

We prepared 10 different types of vibrotactile stimulations using different presentation speeds for the IVS and SOVS. The presenting speeds of both the IVS and SOVS were multiplied by 0.1, 0.5, 1.0 (measured vibration), 1.5, and 2.0, and were edited for a presentation time of 20 s. The experiment was conducted with twelve participants (21–25 years old, ten male, two female, ten right-handed, two left-handed [17]. The participants provided informed consent before the experiment. The experiment was based on the Declaration of Helsinki and approved by the Ethics Committee of the institution. During the experiment, we had all of the participants focus on a yellow gaze marker (20mm x 20mm) on the floor 380mm away from the footwear device in order to eliminate the influence of visual information and used their dominant hand to hold the handheld device. The order of the stimulations was switched and counterbalanced for each participant and the order of the presenting speeds was randomized.

Each participant was asked to indicate how strongly they felt the intensity of the imaginary ball moving back and forth on a five-point scale. The scale was set as follows:

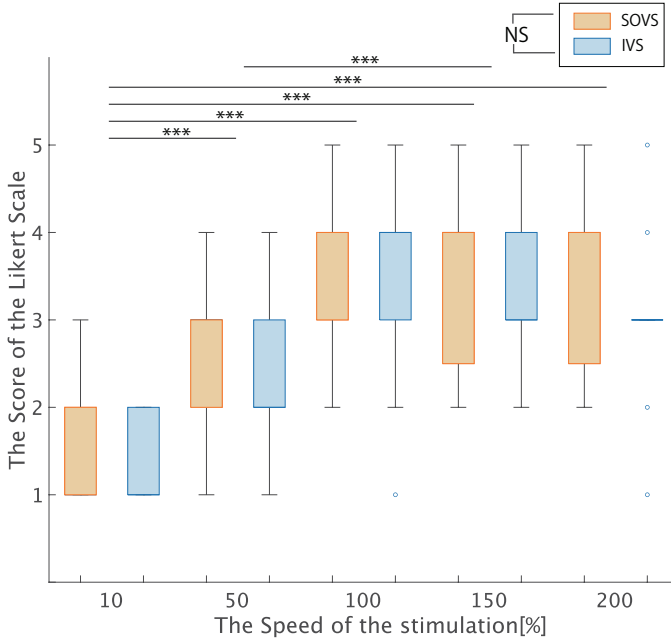


Fig. 7: Results of Experiment 1 (** $p < .001$, ** $p < 0.01$, * $p < .05$)

1. Not Perceivable, 2. Barely Perceivable, 3. Easily Perceivable, 4. Strongly Perceivable, 5. Extremely Strongly Perceivable. Each stimulation was only presented once to measure each participant's intuitive answers. For the data collected, we conducted a two-way repeated measure analysis of variance (ANOVA) after performing an aligned rank transform [18]. The aligned rank transform allowed us to treat the non-parametric data as parametric data, between the answers for the IVS and SOVS with the type of stimulation and the speed as factors. When a significant difference was observed, the Wilcoxon signed-rank test with Bonferroni correction was conducted for post hoc multiple comparisons.

B. Results

The results of the experiment are shown in Fig. 7. For both the IVS and SOVS stimulations, slowing down the stimulation leads to a reduction in the experience of the movement feeling. The movement feeling for the IVS and the SOVS mostly resulted in the same range. In particular, the scores for the IVS resulted in a narrow range of answers in most conditions.

The two-way repeated measure ANOVA showed a significant difference in the speed factor ($F(4, 44) = 32.1$, $p < 0.001$). There were no significant differences in either the stimulation or interaction. The multiple comparisons for the speed indicated there was a significant difference between 0.1x and the other speeds (0.5x: $p < .001$, 1.0x: $p < .001$, 1.5x: $p < .001$, 2.0x: $p < .001$), and also between the answers of 0.5x and 1.5x ($p < .001$) indicating that the movement feeling was significantly decreased when slowing down the stimulation. No significant difference was

observed between IVS and SOVS indicating that SOVS can also induce the movement feeling.

IV. EXPERIMENT 2: THE EFFECT OF ATTENTION

The second experiment aimed to investigate the contribution of the held image of the movement. Naturally, it is difficult to control people's imagination. Thus, we substituted controlling the participant's imagination by interrupting the held image of movement by using an obstacle, in this paper a plastic board, that was installed between the participant's hand and feet. Furthermore, in order to control the focus on the obstacle, we used a gaze marker. In our hypothesis, IVS may induce a tactile apparent motion of the ball moving back and forth between the two devices even when the image is interrupted, whereas when SOVS will only induce a phenomenon similar to tactile apparent motion between the hands and the feet only when the image of movement isn't interrupted.

A. Method

For the obstacle, a white plastic board (300mm x 450mm x 5mm) was set between the handheld device and the footwear device as shown in Fig. 5. We had 12 participants (21–25 years old, nine male, three female, nine right-handed, three left-handed [17]) perform the experiment. For the stimulation, we cut out 10s of the vibration for both the IVS and SOVS. The reason we cut out 10s is that the longer the stimulation, the harder it should become to control the direction of attention. To create a common image of the dribbling movement, each participant was presented with a recorded video taken while measuring the vibration of dribbling. This video was presented with the handheld device in their hands but with no tactile stimulation for 20s before the experiment. Here, through the preliminary experiment, when the obstacle was inserted, certain participants expressed that they could "feel" the imaginary ball reflecting from the obstacle. It was expected that the perception would depend on the participant's recognition of the environment and imagination. Thus, the participants were asked how the tactile apparent motion was felt using two questions. One was "How strong did you feel the ball PIERCE the board in front of you?", and the other question was "How strong did you feel the ball REFLECT on the board in front of you?". The participants were asked to answer both questions on a seven-point Likert scale as follows: 1-Strongly Disagree, 2-Disagree, 3-Somewhat Disagree, 4-Neither Agree nor Disagree, 5-Somewhat Agree, 6-Agree, 7-Strongly Agree.

To control the participant's direction of attention, and when to interrupt the image of the dribbling movement, we put a yellow marker (20mm x 20mm) as a gaze marker on the floor and on the obstacle obscuring the feet. The markers were set 380mm and 280mm away from the participant, respectively, shown in Fig. 8. The marker on the floor was put on the floor slightly further away so the participants

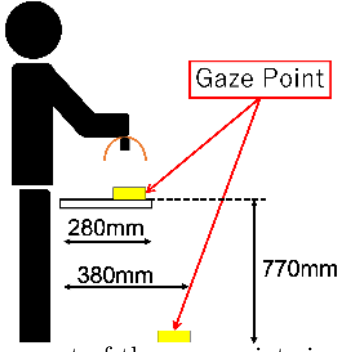


Fig. 8: Placement of the gaze points in experiment 2

were able to see it. Each stimulation was presented once to collect their intuitive answers. All participants conducted both the SOVS and IVS stimulations with all attention conditions. The orders of SOVS and IVS and of the direction of attention were counter-balanced (one-quarter of the participants started with SOVS or IVS with the attention on the obstacle or the floor). As a break between the SOVS and IVS conditions, we had the participants take off both devices and stretch their bodies out for 10s to "wash out" the effects of the prior stimulation. For the data collected, the Wilcoxon signed-rank test in the scores of the piercing or reflection feeling for each condition of the attention with Bonferroni correction.

B. Results

The results of Experiment 2 are shown in Fig. 9 and 10, which show boxplots of the scores of the Likert scale for each IVS and SOVS, respectively. The size of each point represents how many participants answered the same answer. The horizontal axis represents the score for piercing and the vertical axis represents the score for reflection. The grid depicted in the graph represents the boundary of the score where the participants felt either piercing or reflection feeling. The Wilcoxon signed-rank tests in the scores for piercing or reflection were conducted against the score of "4" for each condition of the attention in the IVS or SOVS stimulation. Here, the Bonferroni correction was performed for multiple comparisons. The results for piercing showed a significant difference when gazing at the floor for both IVS ($p<.001$) and SOVS ($p=.004$). Gazing at the obstacle showed a significant difference when IVS ($p=.006$) was presented but there was no significant difference when SOVS was presented. This indicates that when presenting IVS, most of the scores are significantly higher than the score of "4", which is the boundary, for both conditions of attention, ie., most of the participants felt the piercing feeling strongly. On the other hand, when SOVS was presented, the score of piercing depended on the conditions of the attention. With the attention to the floor, the participants felt the piercing feeling significantly whereas, with the attention to the obstacle, some of the participants were not able to feel the piercing feeling. This indicates that IVS produces a strong tactile apparent

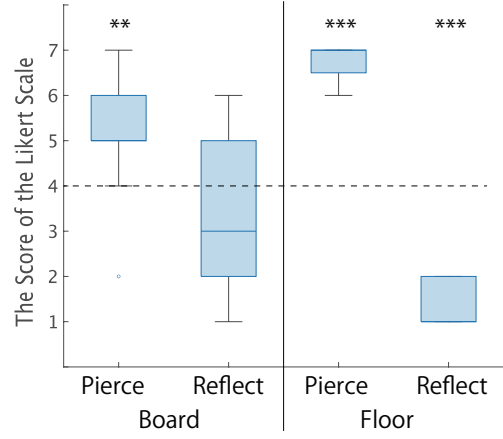


Fig. 9: Results of Experiment 2: IVS(*** $p<.001$, ** $p<0.01$, * $p<.05$)

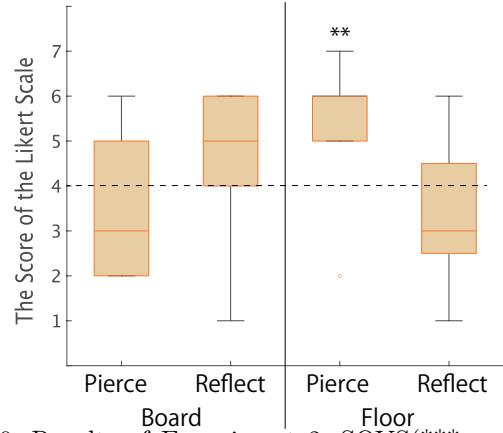


Fig. 10: Results of Experiment 2: SOVS(*** $p<.001$, ** $p<0.01$, * $p<.05$)

motion that can overcome the interruption of the image from the obstacle while SOVS can produce a similar feeling of motion but depending on the image the participant is holding.

The result for reflection is significantly lower than the score of "4" when gazing at the floor and presenting IVS ($p<.001$), indicating that IVS does not induce the feeling of reflection with attention to the floor. There were no significant differences when SOVS was presented with attention to the floor. This indicates that when presenting SOVS and paying attention to the floor, some of the participants felt reflection while the piercing feeling was significantly perceived. When gazing at the obstacle, there were no significant differences for either IVS or SOVS. This indicates that the feeling of reflection is dependent on the individual.

V. DISCUSSION

The results from Experiment 1 showed that the participants could have a feeling that the virtual ball was moving between the hand and foot similar to a tactile apparent motion under both IVS and SOVS presented. A possible reason why such a perception was evoked when presenting the SOVS is that the SOVS allows participants to match

the perceived stimulation to the stimulation according to the imagined dribbling movement. In the SOVS, the vibrotactile stimulation to the hand was combined with that to the feet in the IVS, i.e., the same stimulation was given to both the hand and the foot as shown in Fig. 2. Imagining dribbling might induce a perception of the stimulation on the hand or foot as strong or weak. It seems that the score for the movement feeling was relatively high for both stimulations 1.0x and 1.5x. The scores for the IVS and SOVS resulted in a similar range which indicates that the phenomenon similar to the tactile apparent motion is useful for a dribbling experience.

The result for the different speeds supports the aforementioned discussion. The comparison among speeds of the stimulation indicated significant differences between the scores in both SOVS and IVS, and the score for the stimulation with the speed of 0.1x was significantly smaller than those for other speeds. This indicated that participants did not feel the motion in either IVS or SOVS when presented at 0.1x. When the stimulation is slowed down to 0.1x, there is a large time difference between the stimulation and the touch of the hand or the foot with the ball. Low-frequency stimulations have been reported as not being capable of generating a tactile apparent motion in IVS [19]. In the SOVS, participants were given the same stimulation to both the hand and the foot. Therefore, it might have been difficult to match the stimulation to the imagination of dribbling in the SOVS because the participants could easily recognize each stimulation.

The results of Experiment 2 are shown using a scatter diagram representing the piercing and the reflection feelings of the virtual ball against the obstacle between the hand and the foot. Regarding the IVS, under the attention to the floor, participants significantly perceived the piercing feeling but significantly disagreed with the reflection feeling. This supports the conventional illusion of the tactile apparent motion. It is reported that the vibrotactile stimulation with the time difference to the front and back sides of a body with a time difference evoked a piercing feeling of a virtual object being inserted into the body [20]. Under attention to the obstacle, participants still significantly perceived the piercing feeling but there was no significant tendency in the reflection feeling. This indicated that the attention to the obstacle could influence the perception, but the tactile apparent motion still strongly occurred. The SOVS results showed that attention significantly influences the perception of the piercing feeling. Under the attention to the floor, a piercing feeling was significantly perceived, but under the attention to the obstacle, no significant piercing feeling appeared. This result indicates that the mechanisms of perception are different between the SOVS and IVS. The obstacle interrupts the image of the dribbling movement. The image of the movement that the users hold may be important for evoking the piercing feeling in SOVS. For the reflection feeling, there was no significance under either the

attention to the floor or the obstacle. Therefore, the image of reflection at an obstacle might also be difficult using SOVS, as no significant feeling of reflection was observed.

The current study has several limitations. First, we did not investigate the necessity of similarity between our stimulations and real stimulations. In this experiment, we used the vibrations measured from the floor and the index finger and we discussed the relevance of the image of movement to SOVS. The interface presenting the stimulation may also influence the results, considering that the obstacle significantly affected the piercing feeling in the experiment. Hence, we will need to investigate the influences of the similarities in the stimulations and environment. In addition, in both experiments, all participants had some experience with dribbling a basketball and the effects of the experience were unclarified. The strength of experience may influence the perceptions of SOVS. As noted above, a video of dribbling was presented to the participants. This could be useful in creating a common image consistent with the movements recorded for actual dribbling. We will investigate the effects of this experience in future works. In addition, we instructed the participants to hold a common image by showing a video dribbling and controlled the direction of attention by creating a gaze point. The effects of the direction of attention were demonstrated. For future research, we plan to investigate different methods for controlling attention by using a visual cue and for interrupting the image of movement using multi-modal stimulation.

VI. CONCLUSION

In this study, we hypothesized that presenting Single Overlapper Vibrotactile Stimulation (SOVS) can induce a feeling similar to the tactile apparent motion with IVS by using an image of the movement. To substantiate our hypothesis, we developed a pseudo-dribbling system and conducted two experiments to examine the influences of realism on stimulation and imagination. From the results of the two experiments, we were able to see the possibilities of SOVS and the contributions from the realism of stimulation and imagination. From Experiment 1, we can see that SOVS has the potential to generate a sensation similar to the tactile apparent motion produced by IVS, but the phenomenon disappears when the stimulation is too slow. From Experiment 2, we can see that SOVS is also capable of generating a sensation similar to the tactile apparent motion even when there is an obstacle between the two devices if the user is holding an image of the movement.

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